

ASME B31.3 Pipe Wall Thickness Report

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Input Parameters

Nominal Pipe Size	NPS 2
Outside Diameter	60.3 mm
Design Pressure	1.034 MPa
Design Temperature	260 °C
Material	A53 Grade B
Weld Type	Seamless
Material Type	Ferritic
Corrosion Allowance	1.5875 mm
Threaded Connection	Yes
Mill Tolerance	12.5%

Calculation Details

Pressure Design Thickness (t_dis)	0.238 mm
Corrosion Allowance (CA)	1.587 mm
Thread Depth (TD)	1.913 mm
Allowable Stress (S)	130.31 MPa
Weld Joint Efficiency (E)	1.000
Y Coefficient (Y)	0.400

Results Summary

Required Thickness (t_req)	3.739 mm
Minimum Thickness (t_min)	4.273 mm
Thin-Wall Condition	OK
Overall Status	PASS

ASME B36.10M Schedule Thicknesses

Schedule	Thickness (mm)	Status
5	2.77	✗ Inadequate
10	3.05	✗ Inadequate
40	3.91	✗ Inadequate
80	5.54	✓ Adequate
120	7.01	✓ Adequate
160	8.74	✓ Adequate
XXS	11.07	✓ Adequate